

Instrumeteor: Authoring tool for Guitar Performance Video

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Figure 1: Example of video shots generated by Instrumeteor. Multiple effects informed by the survey are implemented.

ABSTRACT

To show off their playing, musicians publish musical performance videos on streaming services. In order to find out typical characteristics of guitar performance videos, we carried out a quantitative survey of guitar performance videos. Then, we discuss key problems of creating effects informed by the survey. According to the discussion, authoring videos with typical effects takes a long time even for experienced users because they typically need to combine multiple video tracks (e.g., lyrics and videos shot from multiple angles) into a single track. They need to synchronize all tracks with the musical piece and set transitions between them at the right timing, aware of the musical structure. This paper presents Instrumeteor, an authoring tool for musical performance videos. First, it automatically analyzes the musical structure in the tracks to align them on a single timeline. Second, it implements typical video effects informed by the survey. In this way, our tool reduces manual work and unleashes the musicians' creativity.

CCS CONCEPTS

• **Human-centered computing** → *Human computer interaction (HCI); Web-based interaction; Interactive systems and tools.*

KEYWORDS

creativity support tool, video authoring tool, musical instrument performance

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1 INTRODUCTION

Musicians publish musical performance videos on streaming services to show off their playing. In Japan, a lot of performance videos covering the original songs have been published. These Japanese videos tend to have various video effects. This is because the culture of cover creation is advanced in Japan. For example, Nico Nico Douga (<http://www.nicovideo.jp>), the most popular Japanese video-sharing service, the number of tags that are attached to cover performance videos has exceeded 250,000 posts as of January 2021. In addition, according to excerpts of the huge reuse network in the VOCALOID community on the Nico Nico Douga [6], performing instruments video that is cover creation using the musical composition and vocal track of the original song has a lot of views. In this way, musical performance videos are contents that a lot of people, including Japanese, watch and post. Therefore, authoring performance videos is important for these people. However, the existing tool such as Adobe Premiere Pro (<https://www.adobe.com/products/premiere.html>) does not specialize on a specific scene such as performance videos.

In this paper, we propose Instrumeteor, an authoring tool for musical performance videos. Since effects and authoring methods are different depending on the type of instrument, we focus on the guitar performance videos which are popular among all types of performance videos. First, we carry out a quantitative survey on Japanese guitar performance videos which tend to have various effects. Then, we find out characteristics of existing guitar performance videos. We discuss key problems of creating the effects informed by the survey. Next, we propose Instrumeteor, an authoring tool for musical performance videos, which automatically analyzes the musical structure in the tracks to align them on a single timeline. In addition, it implements typical effects informed by the survey such as transition effects and lyrics animation at the

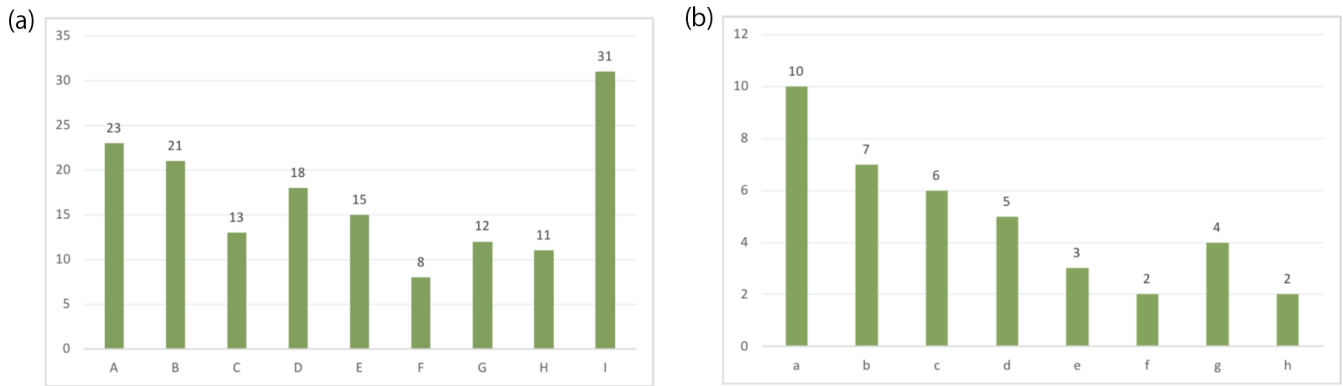


Figure 2: The survey results of 100 guitar performance videos. (a) Results of the quantitative survey on video effects, (I) is no effects. (b) Result of the quantitative survey on video transitions.

right timing (Figure 1). In this way, this tool reduces manual work and unleashes the musicians' creativity. This tool solves the key problems of authoring performance videos with existing tools.

We surveyed Japanese performance videos which tend to have various effects because of the advanced culture of cover creation. However, the usefulness of our tool is not limited to Japan. This tool is beneficial to musicians all over the world. It will lead users to create attractive performance videos that have various effects and unleash their creativity. Therefore, it will promote the improvement of the quality of performance videos around the world.

2 RELATED WORK

Various tools for authoring videos have been proposed. For example, URL2Video [3] allows automatic video creation from web pages. Dynamic manga [2] is a system that creates videos from information on cartoon pages. Then, authoring videos processes differ depending on video genres. Existing systems for authoring videos focus on conversation videos [8][4], interview videos [1], and commentary videos [10]. In this paper, we focus on musical performance videos.

In addition, various systems on music video have been proposed. DanceReProducer [9] is an N-th order cover creation system that automatically generates a dance video clip appropriate to a given piece of music by segmenting and concatenating existing dance video clips. Tsuchida et al. presented an interactive authoring system for multi-view dance videos [11]. The system allows users to automatically edit dance-performance videos taken from multiple viewpoints into a more attractive and sophisticated dance video. However, these systems use limited information in dance videos, and information in musical performance videos is not supported.

Songle [5] is a public web service that automatically estimates musical elements such as musical structure, hierarchical beat structure, melody line, and chords using music-understanding technologies based on signal processing. TextAlive [7] is a graphical tool that allows interactive editing of kinetic typography videos in which lyrics or transcripts are animated in synchrony with the corresponding music or speech. Our tool uses Songle API (Application

Programming Interface) and TextAlive app API to get the information of the musical elements.

3 TARGET APPLICATIONS: GUITAR PERFORMANCE VIDEO AUTHORIZING

This section introduces the result of quantitative survey on the typical video effects of guitar performance videos and discusses key problems of video authoring with existing tools. Then, we determine the design direction of our tool to support authoring performance videos. For the analysis of this section, we referred to our experience of authoring performance videos.

3.1 Characteristics of Guitar Performance Videos

First, in order to find out and analyze the typical characteristics of guitar performance videos, we collected the top 100 most viewed Japanese guitar performances videos. These videos were posted on YouTube from January 1, 2018 to November 10, 2020, using the tag which is often used with performance video postings. Then, there is no duplication of contributors in collected videos to analyze the tendency of more contributors. There were eight types of video effects in the collected videos (Figure 2 a).

- (A): Lyrics animation: Lyrics texts are animated in synchrony with vocal music (Figure 3).

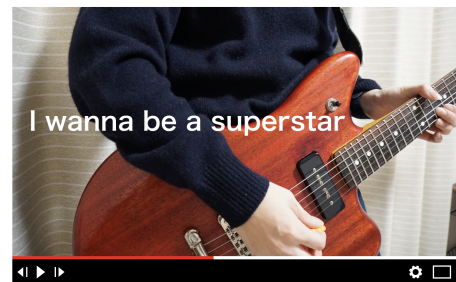


Figure 3: Example of lyrics animation in performance videos.

- (B): Transition effects: When the song melody changes, various types of transition effects are used. The types of transitions are described in detail in the end of this subsection.
- (C): Multi-view switching: The video viewpoint is switched in synchrony with the musical elements (Figure 4). In the chorus section, the viewpoint that captures the entire performance scene is often used. Meanwhile, in the solo section, the viewpoint that captures the fingering scene is often used. In addition, the viewpoints that capture the instrument from various angles are used.



Figure 4: Example of multi-view switching in performance videos. The video viewpoint is switched several times.

- (D): Effect that displays several performance videos at the same time: Different performance shots are partially displayed at the same time (Figure 5).

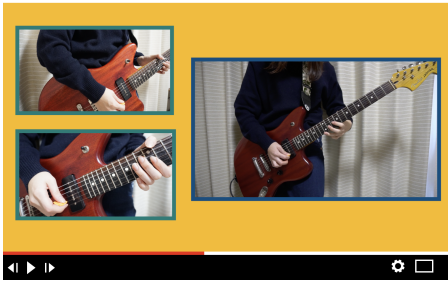


Figure 5: Example of display of multiple videos in performance videos.

- (E): Effect that changes the saturation of a video color when the melody changes: When the melody is quiet, the saturation tends to be low and monochrome (Figure 6).

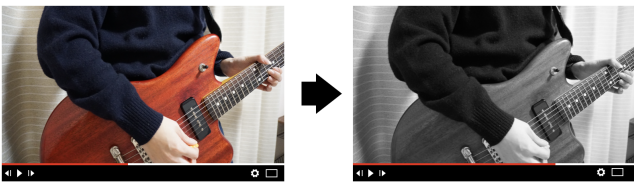


Figure 6: Example of saturation change in performance videos.

- (F): Animation effects such as stamps and shakes: Animation effects are displayed in synchrony with musical elements such as the beat and rhythm of the song (Figure 7 a).
- (G): Effect that displays the music video: The music video for the promotion of the original song is placed in the safe area near the edges of the video (Figure 7 b).

- (H): Effect that displays the score: The score that synchronizes with the performance video is displayed in the video (Figure 7 c).



Figure 7: Example of following characteristics. (a) Animation effects. (b) Display of music video. (c) Display of score.

There are various types of video transitions in the collected performance videos (Figure 2 b). Transitions are used when the melody changes such as the start time of the chorus. The collected videos include the following transitions.

- (a): Cross dissolves: A transition between two shots, with the first shot gradually fading out and the second shot fading in and overlapping.
- (b): Whiteout: A transition where the screen gradually turns white.
- (c): Blackout: A transition where the screen gradually turns black.
- (d): Wipe: A transition where one shot replaces another by travelling from one side of the frame to another.
- (e): Shake: A transition where one shot shakes and replaces another shot.
- (f): Cross Zoom: A transition where the screen zooms in on one clip then zooms out on another.
- (g): A transition where an image is temporarily displayed at the turn of a shot.
- (h): A transition where lyrics are temporarily displayed at the turn of a shot.

3.2 Problems of Guitar Performance Video Authoring

This subsection describes key problems of authoring guitar performance video when editors create effects (A)-(H) with the existing tools.

- (A): When editors create lyrics animation, they need to synchronize lyrics text with the audio manually.
- (B): When editors create transition effects, they need to synchronize transition effects with the melody change manually, aware of the musical structure.
- (C): When editors create multi-view switching, they need to align some recorded video tracks from multiple angles into a single timeline manually. Then, they also need to be aware of the structure and the contents of the song.
- (D): When editors create effects that display several performance videos at the same time, they need to switch the videos in synchrony with the musical elements manually. Then, they also need to be aware of musical sections suitable for displaying multiple videos.
- (E): When editors create effects that change the saturation of a video color, they need to be aware of the change in the melody. They need to find a suitable section to desaturate.

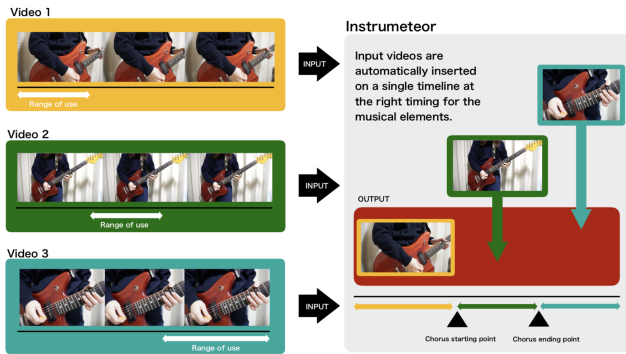


Figure 8: The user selects videos to be displayed for each section. Then, our system automatically inserts the videos on a single timeline at the right timing. These are synchronized with the musical structure.

- (F): When editors create animation effects such as stamps and shakes, editors need to adjust the effects at the right timing for the musical elements with manual work.
- (G): When editors create effects that display the music video, they need to download the music video.
- (H): When editors create effects that display the score, they need to display a part of the score synchronized with the musical progress with manual work.

When editors create these (A)-(H) effects, the existing tools require them to spend much time because they need manual work. In addition, the existing tools are not specialized for performance video authoring. Therefore, it is difficult for editors to trial and error of authoring performance videos.

In this paper, we propose Instrumeteor that solves these problems of the following effects. (A) Lyrics animation, (B) Transition effects, (C) Multi-view switching, (D) Effect that displays several performance videos at the same time, (E) Effect that changes the saturation of a video color when the melody changes, (F) Animation effects such as stamps and shaking, (G) Effect that displays the music video.

4 INSTRUMETEOR

4.1 User Interaction

Musicians record performance videos from multiple viewpoints that capture the entire performance scene, fingering scene and multiple angles of the instrument. At the same time, they record the guitar sound source. They input these recorded videos, recorded sound sources, and song sources to our tool.

- (A): To implement lyrics animation, the existing tools require users to synchronize lyrics text with the audio manually. In our tool, lyrics animation synchronized with the audio is automatically generated, using lyrics information from TextAlive.
- (B): To implement transition effects, the existing tools require users to synchronize transition effects with the melody change manually, aware of the musical structure. Our tool uses the musical section information from Songle such as

information on the start time and end time of each chorus section. The typical transitions informed by the survey are automatically generated at right timing. In addition, users can modify the type of transition depending on their preference.

- (C), (D): To implement multi-view switching and effect that displays some performance videos at the same time, the existing tools require users to do detailed manual work to align multiple video tracks into a single timeline. Our tool uses musical section information from Songle. For example, the user selects videos that he wants to display in the section before the chorus section, between the chorus section and after the chorus section. Then, selected videos are automatically inserted on a single timeline at the right timing that is synchronized with the musical structure (Figure 8). In addition, if a user chooses to display multiple performance videos at the same time, the selected videos are automatically aligned with a single timeline.
- (E): To implement saturation change, editors need to be aware of the change in melody. Our tool automatically generates saturation change synchronized with melody changes using music section information from Songle.
- (F): To implement animation effects such as stamps and shake, the existing tools require them to adjust the effects with the synchronized timing for the musical elements manually. In our tool, animation effects synchronized with the musical elements are automatically generated, using beat information from Songle.
- (G): Display of music videos is realized by directly embedding the video on YouTube in the web page.
- (H): By these authoring functions, users can easily create attractive performance video that is synchronized with the musical elements.

Figure 1 shows example of video shots generated by Instrumeteor. Figure 1 a shows (A) Lyrics animation, (F) Stamps animation, (G) Display of music videos. Figure 1 b also shows (E) Saturation change. Then, (C) the viewpoint changes between Figure 1 a and Figure 1 b. Figure 1 c shows (D) Display of multiple videos.

4.2 Implementation

Instrumeteor is a web application. It uses Songle API and TextAlive app API to get the information of the musical elements. It also uses P5.js to implement effects such as video transition and stamp animation.

In TextAlive, the song and lyrics are automatically synchronized. In Songle, it is processed detection of the musical structure (actually, it is processed in the Songle linked with TextAlive). Our tool implements multi-view switching and multiple videos display using the information of the music structure. Then, it implements randomly the typical transitions informed by the survey. In addition, it implements the lyrics animation synchronized with the song using the lyrics information from TextAlive. It also implements animation effects such as stamps and saturation change synchronized with musical elements using the information from Songle.

5 CONCLUSION

We carried out the quantitative survey of Japanese guitar performance videos where the culture of cover creation is advanced. Then, we found out the typical video effects in existing performance videos. We also discussed key problems of creating those effects.

Next, we proposed Instrumeteor, an authoring tool for guitar performance videos that solves those key problems. It automatically analyzes the musical structure in the tracks to align them on a single timeline. Therefore, it reduces manual work for routine tasks. In addition, it implements typical video effects informed by the survey. By using this tool, users can easily create attractive performance videos while unleashing their creativity.

6 FUTURE WORK

We will conduct a user study in order to compare our tool with other existing tools. In addition, we will use optical flow to judge videos that capture intense fingering scenes or the entire performance scenes. Then, we will implement a function that automatically inserts a performance video suitable for the section of the song without the user's selection task. Finally, this paper focuses on guitar performance videos. For future work, we will survey performance videos of other musical instruments.

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